

Tertiary Syphilis

- **Gummatous syphilis:** Characterized by granulomatous lesions (gummas) presenting as nodules, plaques, or ulcers affecting the skin, mucous membranes, and visceral organs.
- **Late neurosyphilis:** Includes a range of neurological manifestations that may occur years after initial infection:
 - Meningeal involvement (chronic meningitis)
 - Cranial nerve dysfunction
 - Meningovascular syphilis (can lead to stroke or myelitis)
 - Parenchymatous neurosyphilis:
 - General paresis (dementia, psychiatric symptoms)
 - Tabes dorsalis (sensory ataxia, lightning pains, areflexia, bladder dysfunction)
- **Cardiovascular syphilis:** Typically manifests decades after infection:
 - Aortic regurgitation
 - Stenosis of coronary ostia
 - Thoracic aortic aneurysm (especially of the ascending aorta)

> **Note:** Neurological involvement in syphilis can occur at any stage—early (e.g., during secondary syphilis) or late (tertiary syphilis).

Laboratory Diagnosis

Demonstration of *Treponema pallidum*

Direct detection methods provide definitive diagnosis of syphilis.

- **Darkfield examination (DFE):**
 - Historically the gold standard for diagnosing primary syphilis via visualization of motile spirochetes in fluid from chancres or erosive skin lesions.
 - Offers immediate results but is labor-intensive, operator-dependent, and prone to false-negative (common) and false-positive (less common) results.

- Not recommended for routine use due to the availability of more sensitive and specific molecular methods.
- **Polymerase chain reaction (PCR):**
 - Preferred method for direct detection, especially for oral, genital, and cutaneous lesions where contamination with non-pathogenic treponemes may occur.
 - Can be performed on lesion exudate, tissue biopsies, cerebrospinal fluid (CSF), and blood (though blood PCR is insensitive).
 - No internationally standardized assay exists; therefore, laboratories must use rigorously validated, quality-controlled PCR protocols.
- **Immunohistochemistry (IHC):**
 - Uses polyclonal antibodies against *T. pallidum* to detect treponemes in fixed tissue sections (skin, mucosa, organ biopsies).
 - Useful in histopathological confirmation but not suitable for routine clinical diagnosis.
- **Hybridization techniques:**
 - Not used in routine diagnostics due to complexity and limited availability.
- **Warthin-Starry (argentic) staining:**
 - Difficult to perform and interpret; low sensitivity and specificity.
 - Rarely useful in clinical practice.
- **Direct fluorescent antibody test (DFA):**
 - Requires specialized equipment, reagents, and training.
 - Not recommended for routine diagnosis.
- **Molecular epidemiological typing:**
 - Techniques include PCR, PCR-RFLP, multilocus sequence typing (MLST), and whole genome sequencing.
 - Limited by the high genetic conservation of *T. pallidum* genome, resulting in low discriminatory power among strains.

Serological Tests for Syphilis (STS)

Serological testing remains essential for the presumptive diagnosis of syphilis.

- **Important limitation:** STS cannot differentiate between venereal syphilis (*T. pallidum* subsp. *pallidum*) and non-venereal treponematoses:
- Yaws (*T. pallidum* subsp. *pertenue*)
- Bejel (endemic syphilis; *T. pallidum* subsp. *endemicum*)
- Pinta (*T. carateum*)
- These organisms are morphologically and antigenically similar. Differentiation relies on clinical presentation, epidemiology, mode of transmission, and genomic sequencing.

Clinical implication: A reactive STS should be managed as syphilis unless there is documented evidence of prior adequate treatment (Grade 1, Level D recommendation).

Non-treponemal tests (NTT)

- Detect antibodies (IgG and IgM) against cardiolipin-lecithin-cholesterol antigen complexes (so-called "reagin" tests).
- Examples:
- Venereal Disease Research Laboratory (VDRL) test
- Rapid Plasma Reagin (RPR) test
- Tolidine Red Unheated Serum Test (TRUST)
- Advantages:
- Inexpensive, simple to perform, and widely available.
- Useful for monitoring disease activity and treatment response (titers correlate with disease activity).
- Seroconversion:
- Typically occurs 10–15 days after appearance of the primary chancre (~6 weeks post-infection).
- Natural history:
- Titers peak 1–2 years after infection.

- Remain detectable at low levels even in late, untreated syphilis.
- Spontaneous seroreversion (conversion to negative) is extremely rare in tertiary syphilis.